

Bluetooth Networks and Security Protocols

Module 33

Duration: 30:01

Video: https://www.youtube.com/watch?v=wzD1F-Knr_E

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Prof Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

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0:00 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering of Guru

0:08 jambheshwar University of Science and Technology Hisar harana I welcome you all to the lecture series of cyber and

0:17 information security in this lecture we will study about Bluetooth networks and security

0:25 protocols which is basically used for data transmission on wireless personal

0:32 area network first of all see the contents which I shall cover after covering brief introduction about

0:41 Bluetooth technology I shall take up Bluetooth layered protocol architecture which will be followed by frame format

0:49 of Bluetooth then I shall discuss versions of Bluetooth and i e standards

0:57 for w pen technology in last I shall take up security architecture of

1:04 Bluetooth technology so now let us start Bluetooth is a wireless technology

1:12 standard for exchanging data over short distances using short wavelength UHF

1:19 radio waves it operates in the ism band from 2.4 to

1:28 2.4856 and mobile devices and building personal area networks in

1:35 1994 Ericson company became interested in connecting its mobile devices to

1:42 other devices without cables together with four other companies IBM Intel Nokia and Toshiba it

1:53 formed a special interest group they developed a wireless standard

2:00 for interconnecting computing and communication devices and accessories

2:06 using short range low power inexpensive Wireless radios this technology got named

2:14 Bluetooth after the 10th Century King Herald Bluetooth who United disant

2:21 Danish tribes into a single Kingdom the idea of this name was proposed in

2:28 1997 by by Jim card who developed a system that allowed mobile phones to

2:35 communicate with computers Bluetooth special interest group SI defined standards for

2:44 manufacturing Bluetooth devices in July 1999 the Bluetooth SI issued a 1,500

2:53 page specification of version 1.0 shortly thereafter the i.e. standards

3:01 3 minutes, 1 second group for wireless personal area network 802.15 adopted the Bluetooth document as

3:10 a basis and started working on it Bluetooth specification SI is for a

3:18 complete system that is from the physical layer to the application layer but i.e.

3:27 802.15 defines only the physical and data link layers the basic unit of a Bluetooth

3:36 system is a piconet as shown in figure which consist of a master node and up to

3:44 seven active slave nodes within a distance of 10 m multiple piconets can exist in the

3:54 same large room and can even be connected via a bridge node as shown in

4:02 figure an interconnected collection of piconets is called a scatternet all communication is between

4:11 the master and the slave direct slave to slave communication is not

4:20 possible after a brief background let us see layer protocol architecture of Bluetooth the Bluetooth standard has

4:29 many protocols grouped loosely into layers the basic Bluetooth protocol

4:36 architecture is shown in figure the bottom layer is the physical radio layer which corresponds fairly

4:45 well to the physical layer in the OSI and 802 models it deals with radio

4:53 transmission and modulation using ISM band from 2.4 to

5:01 5 minutes, 1 second 2485 GHz frequency the baseband layer is somewhat

5:08 analogous to the max sublayer but also includes elements of the physical layer

5:15 it deals with how the master controls time slots and how these slots are grouped into

5:24 frames the radio layer moves the bits from Master to slave or vice

5:30 versa this layer turns the raw bit stream into frames and defines some key

5:38 formats in the simplest form the master in each piconet defines a series of 625

5:46 microsecond time slots with the Master's transmission starting in the even slot and the slave transmission starting in

5:55 the odd slots this is traditional time division multiplexing with a master getting half

6:03 the slots and slaves sharing the other half each frame is transmitted over a

6:12 logical Channel called a link between the master and the slave two kind of

6:19 Link exist the first is the ACL that is asynchronous connectionless link which

6:26 is used for packet switch data available at irregular intervals the other is the SCO that is

6:35 synchronous connection oriented link for realtime data such as Telephone
6:42 Connections this type of channel is allocated a fixed slot in each Direction due to
the time critical
6:51 nature of SEO links frames sent over them are never retransmitted instead forward
error correction can be
7:00 used to provide High reliability next layer link manager handles the establishment
of logical
7:09 channels between devices including power management authentication and quality
of
7:19 service The Logical link control adaptation protocol often called l2cap Shields the
upper layers from the
7:28 details of transmission the l2cap layer has three major functions first it accepts
packets
7:38 of up to 64 KB from the upper layers and breaks them into frames for transmission
7:46 at the far end the frames are reassembled into packets again second it handles the
multiplexing
7:56 and demultiplexing of multiple packets sources third l2cap handles the quality
8:04 of service requirements as the name suggest the audio and control protocols deal
with
8:13 the audio and control respectively the applications can get at them directly without
having to go
8:22 through the L2 cap protocol the next layer contains a mix of different prot
8:29 protocols the RFC radio frequency communication is the protocol that emulates the
standard serial Port found
8:39 on PCs for connecting the keyboard mouse and modem among other devices the
8:47 telephon protocol is a realtime protocol used for the three speech oriented
8:55 protocols it also manages call setup and termination finally the service
9:03 Discovery protocol is used to locate services within the network the top layer is
where the
9:11 applications and profiles are located they make use of the protocols in lower
9:18 layers to get their work done so this was layered protocol architecture of
9:25 Bluetooth now let us see frame format of Bluetooth the frame format of Bluetooth
9:33 is shown in figure it begins with an SS code that usually identifies the master
9:41 so that slaves within radio range of two masters can tell which traffic is for
9:48 them next comes a 54 bit header containing typical Max sublayer
9:56 Fields then comes the data field of up to 2744 bits for a five slot
10:04 transmission for a single time slot the format is same except that the data field is
240

10:14 bits let us take a quick look at details of the header the address field
10:21 identifies which of the eight active devices the frame is intended for the
10:29 type type field identifies the frame type whether it is ACL SE Pole or null
10:38 type it also tells the type of error correction used in the data field and
10:46 how many slots long the frame is the flow bit is for flow control
10:54 which is asserted by a slave when its buffer is full and cannot receive any more
11:01 11 minutes, 1 second data the acknowledgement bit is used to piggy back an AK
onto a
11:09 frame the sequence bit is used to number the frames to detect retransmissions the
protocol is stop and
11:18 wait so one bit is enough then comes the 8bit header checkm
11:26 the entire 8bit header is repeated three times to form the 54 bit header shown in
11:35 figure on the receiving side a simple circuit examines all three copies of
11:42 each bit if all three are same the bit is accepted if not the majority opinion
11:51 wins Bluetooth SI has defined versions for Bluetooth and i e has has defined
12:00 standards for wireless pan let us first see versions defined by Si the versions
12:08 were formalized by the Bluetooth special interest group SI the SI was formally
announced on 20th
12:18 May 1998 today it has a membership of over 20,000 companies
12:25 worldwide it was initially established by Ericson IBM Intel Toshiba and Nokia
12:34 and later it was joined by many other companies all versions of the Bluetooth
12:42 standard support downward compatibility let us see each version one by
12:49 one Bluetooth v1.0 and v1.0b these are basic most Bluetooth
12:58 which had many problems and manufacturers had difficulty making their products
13:06 interoperable Bluetooth v1.1 it was ratified as I E standard
13:15 802.15 2002 many errors found in the version 1.0b
13:22 specifications were fixed in this version it added possibility of known and cryp red
channels along with
13:31 received signal strength indicator Bluetooth v1.2 this version supports faster
13:40 connection and Discovery protocols it supports adaptive frequency hoping
13:47 spread Spectrum which improves resistance to radio frequency interference by
avoiding the use of
13:55 crowded frequencies in the hoping sequence it supports higher transmission
speeds
14:02 in practice up to 721 kilobits per second than in
14:10 v1.1 it was ratified as IA standard 82.1
14:17 15.12 2005 Bluetooth v2.0 plus
14:25 EDR this version of the Bluetooth core specification was released in

14:32 2004 the main difference is the introduction of an enhanced data rate
14:38 EDR for faster data transfer the nominal rate of EDR is about 3 megabits per
14:46 second although the Practical data transfer rate is 2.1 megabits per second
14:54 EDR uses a combination of gfsk and phase shift Key
15:01 15 minutes, 1 second modulation Bluetooth 2.1 plus EDR Bluetooth core
specification version 2.1
15:10 plus EDR was adopted by the Bluetooth SIG on 26 July
15:17 2007 the headline feature of v2.1 is secure simple pairing
15:25 SSP this improves the pairing experience for Bluetooth devices while increasing
15:32 the use and strength of security Bluetooth v3.0 plus
15:41 HS this version was adopted by the Bluetooth SIG on 21st April
15:48 2009 Bluetooth v3.0 plus HS that is high speed provides theoretical data
15:56 transfer speed of up to 20 4 megabits per second though not over the Bluetooth
16:04 Link itself instead the Bluetooth Link is used for negotiation and
16:11 establishment and the high data rate Traffic is carried over a 802.11
16:19 link Bluetooth v4.0 the Bluetooth SIG completed the
16:26 Bluetooth core specification version 4.0 called bluetooth smart and has been
16:33 adopted on 30th June 2010 it includes classic Bluetooth
16:40 Bluetooth highspeed and Bluetooth low energy LE protocols Bluetooth LE
previously known
16:49 as BLE is a subset of Bluetooth v4.0 with an entirely new protocol stack
16:57 for Rapid buildup of simple links Bluetooth
17:03 v4.1 the Bluetooth SIG announced formal adoption of the Bluetooth v4.1
17:11 specification on 4th December 2013 this specification is an
17:17 incremental software update to Bluetooth specification v4.0 and not a hardware
17:26 update features include increased coexistence support for LTE bulk data
17:33 transfer exchange rates and Aid developer Innovation by allowing devices
17:40 to support multiple roles simultaneously Bluetooth
17:47 v4.2 Bluetooth v4.2 was released on December 2
17:54 2014 it introduces some key features such as data length extension that
18:00 require a hardware update but some older Bluetooth Hardware may receive some
18:07 Bluetooth v4.2 features such as privacy updates via firmware update
18:15 also now after Bluetooth version let us see i.e standards for wireless personal
18:22 area network IEEE 802.15 is a working group of The
18:29 Institute of electrical and electronic Engineers i.e which specifies Wireless
18:37 personal area network standards now we will take up each standard one by one as
shown in

18:46 figure i e 802.15.1 WPAN

18:53 Bluetooth this standard is based on Bluetooth technology it defines physical layer
phy and media access

19:02 control MAC specification for wireless connectivity with fixed portable and
19:10 moving devices within or entering personal operating
19:16 space standards were issued in 2002 and
19:22 2005 IEEE 802.15.2 coexist
19:30 this standard addresses the coexistence of Wireless personal area networks WPAN
with other wireless devices

19:40 operating in unlicensed frequency bands such as Wireless local area networks WLAN
19:48 LAN the IEEE 802.15 point-to-point 2003 standard was published in
19:57 2003 and task group two went into hibernation IEEE 802.1
20:06 15.3 High rate WPAN IEEE
20:12 802.15.3 2003 is a MAC and PHY standard
20:19 for high rate of 11 to 55 Mbps in Wireless personal area
20:26 networks this standard has three standards IEEE 802.15.3 a
20:36 802.15.3 B 2006 and 802.1
20:44 15.3 C 2009 next is IEEE
20:51 802.15.4 low rate WPAN IEEE 802.15
20:59 .4 2003 low rate WPAN deals with a low data rate but very long battery life and
21:08 very low complexity the standard defines both the physical and data link layers of
the OSI
21:17 model this standard has two versions first is WPAN low rate alternative PHY
21:24 4A formally called IEEE 802.15.4 A2
21:34 2007 second is revision and enhancement 4B formally called as IEEE
21:43 802.15.4 2006 IEEE 802.15.5 Mesh
21:52 networking it provides the architectural framework enabling W devices to promote
22:01 22 minutes, 1 second interoperable stable and scalable Wireless Mesh networking
this standard is composed of
22:10 two parts low rate WPAN meshless and high rate WPAN mesh
22:18 networks the low rate mesh is built on IEEE 802.15.4
22:27 2006 meanwhile the high rate mesh utilizes IEEE 802.1
22:35 15.3 3B MAC next is IEEE 802.1
22:41 15.6 body area networks in December 2011 the IEEE 802.1
22:51 15.6 task group approved a draft of a standard for body area network
22:59 BAN Technologies focus is on devices in or around the human body to serve a
variety
23:08 of applications including medical consumer electronics and personal

23:15 entertainment other standards are i.e. 802.15.7 that is visible light communication i.e. 802.15.8 that is peer aware Communications i.e. 802.15.9 that is Key Management protocol and last is i.e. 802.15.10 that is layer 2 routing after versions and standards we now move on to security architecture of Bluetooth three basic security services are specified in the Bluetooth standard authentication it is for verifying the identity of communicating devices based on their Bluetooth device address Bluetooth does not provide native user authentication second is confidentiality it is for preventing information compromise caused by EES dropping by ensuring that only authorized devices can access and view transmitting data third is authorization it is for allowing the control of resources by ensuring that a device is authorized to use a service before permitting it to do so Bluetooth does not address other security services such as audit integrity and non repudiation if such services are needed they should be provided through additional means the family of Bluetooth BR that is basic rate EDR that is enhanced data rate HS that is highspeed specifications defines four security modes each Bluetooth device must operate in one of these modes let us see these modes one by one security mode one in this mode devices are considered non-secure security functionality like authentication and encryption is never initiated leaving the device and connections susceptible to attackers in effect Bluetooth devices in this mode are indiscriminate and do not employ any mechanisms to prevent other bluetooth enabled devices from establishing connections however if a Remote device initiates security such as a pairing authentication or encryption request a security mode one device will participate all v2.0 and earlier devices can support security mode 1 and v2.1 and later devices can use security mode one for backward compatibility with other devices but NIST recommends never use security mode one for Bluetooth devices second is security mode two in this mode security procedures may be initiated after link establishment but before logical Channel establishment for this security mode a local security manager as specified in the Bluetooth architecture controls access to specific de Services the centralized security manager maintains policies for Access Control and interfaces with other protocols and device users security mode three this mode is the link level enforced security mode in

27:40 which a Bluetooth device initiates security procedures before the physical link is fully
27:48 established Bluetooth devices operating in security mode 3 mandate authentication and encryption for all
27:57 connections to and from the device all v2.0 and earlier devices can support
28:05 security mode 3 but v2.1 and later devices can only support it for backward
28:14 compatibility purposes last is security mode 4 similar to security mode 2 security
28:24 mode 4 introduced in Bluetooth v2.1 plus EDR is a service level and fr
28:32 security mode in which security procedures are initiated after physical
28:39 and logical link setup security mode 4 uses secure simple pairing SSP in which
28:50 elliptic curve Dey Helman key agreement replaces Legacy key agreement for link
28:58 key generation however the device authentication and encryption algorithm
29:05 are identical to the algorithm in Bluetooth v2.0 plus EDR and earlier
29:14 versions so friends this was Bluetooth networks and security protocols finally
29:23 summarizing we learned about Bluetooth lead protocol architecture which was followed by frame format of Bluetooth
29:32 then we discussed versions and it standards for Bluetooth technology in
29:40 last we covered security architecture of Bluetooth technology hope the concepts explained
29:48 in this lecture were understandable and helpful hope to see you in next lecture
29:56 till then goodbye enjoy the day thank you